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Course: ITAI-1371-Intro to Machine Learning

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Module 03 Lab Reflective Journal

ML Workflow & Learning Types

Section 1: Learning the ML Workflow and Overcoming Fear of Code

When I first opened the Module 3 Lab notebook, the machine learning workflow seemed completely overwhelming. Seeing 25 cells filled with long Python code made me nervous. At first, I just clicked the "Run" button on Google Colab to get through the steps quickly, without understanding the code lines.

However, as I ran the cells one by one, my thinking changed. I realized I do not need to memorize hard functions like `train_test_split()` or `classification_report()`. Instead, I just need to understand how data moves. I began to see each code block as a small "black box" inside a continuous lifecycle. The tools I learned in Lab 2, like NumPy lists and Seaborn charts, fit perfectly here as the gears that move the data. This took away my anxiety and allowed me to focus on data logic instead of memorizing text.

Section 2: Breakthroughs in Understanding Learning Paradigms

Distinguishing supervised from unsupervised learning clicked when I realized it is simply about whether you have an "answer key" during training. In our wine project, splitting the data into a feature matrix (X) and a target label (y) created a supervised system. The computer calculates using the `.fit()` command, makes a guess, checks the human answer key, and fixes its weights.

This is very different from reinforcement learning, which is now my favorite concept. Here, a computer agent learns by actively trying things in a changing world. It uses a loop of rewards and penalties, just like a human learning a video game. I realized that machine learning is not about writing strict rules, but about creating an environment where a computer teaches itself through trial and error.

Section 3: The Geometry of Mistakes and Why We Sacrifice Data

I initially struggled to understand why we hide 20% of our data in a test set. But looking at our actual scores—where Logistic Regression got 88.9% and the Decision Tree got 83.3%—taught me about overfitting. Without a hidden test set, a model will just cheat by memorizing the training questions instead of learning true chemical patterns.

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This became visually clear when I looked at the Confusion Matrix. Global accuracy scores can lie, but the confusion matrix grid showed me exactly *how* the model failed by plotting wrong guesses outside the main diagonal line.

I also learned that data size changes our strategy. For a small 178-row dataset, a 99/1 split would only leave 2 test questions, which is a bad exam. To fix this, I learned about 5-Fold Cross-Validation. By rotating the data through 5 rounds, every row turns into a test question once and for a training student 4 times. This rotating system lets a tiny dataset act much bigger than it is. It proves we do not need the most complex model in the world; we just need a practical model that is sized correctly to do its job.